

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

Marks

Q.1 Solve any Two of the following

- A) An insurance company insured 2000 scooter drivers, 4000 car drivers and 6000 truck drivers. The probability of accident are 0.01, 0.03 and 0.15 for scooter driver, car driver and truck driver respectively. One of the insured person meets with an accident. What is the Probability that he is a scooter driver ? 6
- B) Explain Addition theorem and Multiplication theorem of probability with example 6
- C) Prove the following properties of probability 6
- i) $P(A' \cap B) = P(B) - P(A \cap B)$
 - ii) $P(A') = 1 - P(A)$

Q.2 Solve any Two of the following.

- A) Define random variable. 6

Suppose a random variable X has the following probability distribution function,

X	1	3	4	7	9	10	14	18
P(X)	0.11	0.07	0.13	0.28	0.18	0.05	0.12	0.06

then find the probability that:

- (a) X is at least 10
 - (b) X is more than 10
 - (c) X is less than 4
 - (d) X is at least 4 and at most 9
- B) In a recent little league softball game, each player went to bat 4 times. The number of hits made by each player is described by the following probability distribution. 6

Number of hits, x	0	1	2	3	4
Probability, P(x)	0.10	0.20	0.30	0.25	0.15

What is the mean, standard deviation of the probability distribution? Find $P(X \leq 2)$ and $P(X \geq 2)$

- C) Explain binomial and Normal distribution 6

Q. 3 Solve Any Two of the following.

- A) What is correlation? Explain different types of correlation. 6

- B) Write two properties of Karl Pearson's correlation coefficient and calculate it for the following information $n=10$, $\Sigma x=40$, $\Sigma x^2=190$, $\Sigma y^2=200$, $\Sigma x.y=150$, $\Sigma y=40$
Comment on type of correlation 6

- C) Write two properties of Spearman's rank correlation coefficient. Two participants in a beauty contest are ranked by two judges as follows: 6

X	1	6	5	10	3	2	4	9	7	8
Y	6	4	9	8	1	2	3	10	5	7

Q.4 Solve Any Two of the following.

- A) The equation of two regression lines obtained in a correlation analysis are $4x-5y+33=0$ and $20x-9y-107=0$. If the variance of y is 16 then find 6

- Mean values of x and y
- The correlation coefficient between x and y
- The variance of x

- B) What is linear regression? Write working rule to find lines of regression 6

- C) Obtain lines of regression for the following data 6

X	6	2	10	4	8
Y	9	11	5	8	7

Find y for $x=5$ and x for $y=10$

Q. 5 Solve Any Two of the following.

- A) What is hypothesis testing? Explain the elements of hypothesis testing 6

- B) Explain: 6

- Large Sample Estimation of a Population Mean
- Small Sample Estimation of a Population Mean
- Large Sample Estimation of a Population Proportion

- C) It is hoped that a newly developed pain reliever will more quickly produce perceptible reduction in pain to patients after minor surgeries than a standard pain reliever. The standard pain reliever is known to bring relief in an average of 3.5 minutes with standard deviation 2.1 minutes. To test whether the new pain reliever works more quickly than the standard one, 50 patients with minor surgeries were given the new pain reliever and their times to relief were recorded. The experiment yielded sample mean $\bar{x}=3.1$ minutes and sample standard deviation $s=1.5$ minutes. Is there sufficient evidence in the sample to indicate, at the 5% level of significance, that the newly developed pain reliever does deliver perceptible relief more quickly? 6

*** End ***